

An Integrated Framework for Explaining Innovation and Product Development Strategies Based on the Innovation Matrix, Ansoff Matrix, and Kano Model

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Abstract

In today's dynamic business environment, organizations must continuously rethink their innovation and product development strategies in order to maintain competitive advantage and respond effectively to market changes. The accelerating pace of technological change, the diversity of customer needs, and the growing complexity of strategic decisions have doubled the need for integrated frameworks to explain and guide these processes. The present study aims to propose an integrated framework for explaining innovation and product development strategies. In this research, the Innovation Matrix, the Ansoff Matrix, and the Kano Model are brought together as three complementary models that, when combined, allow for a more coherent understanding of the relationships among product, technology, market, and customer-need dimensions. Among these, the Kano Model provides a basis for explaining customer expectations and preferences, while the Innovation Matrix and the Ansoff Matrix provide the ground for analyzing different product-development situations and market orientations. On this basis, various product and technology conditions are reformulated within a conceptual framework so that the relationship among product characteristics, market conditions, and innovation pathways can be interpreted in an integrated way. The approach of the present research is qualitative and based on conceptual analysis. By integrating the explanatory capacities of the three aforementioned models, an integrated framework is presented that can be used to identify and explain innovation and product development strategies suited to the situation. The output of the research is an integrated matrix that, by simultaneously considering the dimensions of product, technology, and market, allows for a better interpretation of the fit between customer needs and the organization's strategic orientations. Furthermore, this framework can serve as a basis for understanding and adapting innovation and product development strategies under various organizational conditions.

Keywords: *Innovation Matrix; Ansoff Matrix; Kano Model; Innovation Strategy; Product Development; Integrated Framework.*

1. Introduction

In the current era, organizations operate in an environment characterized by rapid technological change, intense competition, shortened product life cycles, and continuously shifting customer needs and expectations. Under such conditions, the ability to choose an appropriate strategy for innovation and product development has become one of the key factors of success and competitive advantage, particularly for knowledge-based and technology-driven firms. Consequently, organizations need frameworks that can facilitate the decision-making process of selecting, evaluating, and prioritizing innovation and product development strategies, thereby increasing the likelihood of new-product success.

One of the best-known tools for formulating growth strategies is the Ansoff Matrix, which introduces four main strategies, market penetration, market development, product development, and diversification for

achieving organizational growth (Ansoff, 1957). In contrast, the Innovation Matrix focuses on the two dimensions of product and technology, enabling the classification and analysis of various types of innovation, including incremental, radical, architectural, and disruptive innovation, and helping managers choose an appropriate innovation path (Chen & Chen, 2023; Rothaermel, 2024). Despite the widespread use of these two tools in the fields of strategic management and innovation management, a review of the literature shows that no comprehensive study has yet integrated these two matrices into a unified framework for simultaneously analyzing growth, innovation, and product development strategies.

On the other hand, the Kano Model, as one of the best-known tools for analyzing customer needs and satisfaction, classifies product and service characteristics into three main groups: Must-be, One-dimensional, and Attractive. This model has so far been used mainly in the field of quality management, product design, and improving customer satisfaction, whereas its potential for analyzing strategic innovation and product-development decisions has received less attention (Shahin, 2014). Attractive features are capabilities that customers do not expect, but whose provision generates striking satisfaction and competitive advantage; accordingly, these features may be more closely linked to emerging products and technologies, whereas Must-be and One-dimensional features are more closely associated with existing products and technologies. Nevertheless, the role of the Kano Model in distinguishing between existing and new technologies, and likewise in analyzing the technology life cycle and selecting innovation strategies, has not yet been systematically examined.

In addition, in the strategic decision-making process, indicators such as market impact, sales growth, market development, market share, and customer acceptance are considered among the most important criteria for evaluating the success of innovation strategies (Wei, 2025). However, most studies have examined these indicators independently, and so far no framework has been presented that can analyze these criteria together with the type of technology, product status, and growth strategies in an integrated manner. This has left managers facing ambiguity and uncertainty in selecting the most suitable innovation and product development strategy.

Accordingly, the main gap addressed by the present study is the absence of a comprehensive conceptual framework that integrates the Ansoff Matrix, the Innovation Matrix, the Kano Model, and market indicators for analyzing and selecting innovation and product development strategies. This research adopts a qualitative approach and seeks, by integrating two important strategic tools, the Ansoff Matrix and the Innovation Matrix and drawing on the Kano Model, to present a conceptual framework that increases the precision of identifying, analyzing, and evaluating innovation and product development strategies. In this framework, the Kano Model is used to classify product characteristics based on their status (new or existing) and the type of technology (new or existing), so that customer needs and expectations can be identified with greater precision and taken into account in the decision-making process. Market indicators are likewise considered as one of the dimensions for evaluating strategies within the proposed framework, so that the market, technology, and product dimensions can be analyzed simultaneously.

The significance of this research can be explained from both a theoretical and a practical perspective. From a theoretical standpoint, by presenting an integrated framework, this study reduces the existing gap in the

innovation-management and product-development literature regarding the integration of strategic tools, thereby paving the way for future studies. From a practical standpoint, the proposed framework can help managers and decision-makers in selecting a suitable innovation strategy, improving new-product development, upgrading existing products, allocating resources optimally, and reducing decision-making risk.

Given the issues raised above, the main research question is formulated as follows: How can an integrated framework be developed, by combining the Ansoff Matrix, the Innovation Matrix, and the Kano Model, for analyzing and selecting innovation and product development strategies while taking into account the technology, product, and market dimensions?

2. Theoretical Background and Literature Review

Innovation management is one of the most important topics in strategic management, playing a fundamental role in creating competitive advantage, developing new products, and achieving organizational growth and sustainability. In today's competitive environments, organizations are compelled to adopt systematic approaches to innovation management in order to respond to technological change, market shifts, and evolving customer needs. In recent years, concepts such as open innovation, artificial intelligence, and emerging technologies have further increased the importance of innovation management in developing new products and making strategic decisions (Rothaermel, 2024; Haefner et al., 2021).

Based on the innovation-management literature, innovation can be classified, according to the state of technology and market, into four categories: incremental, radical, disruptive, and architectural. Incremental innovation focuses on the continuous improvement of existing products and technologies, whereas radical innovation, by leveraging new technologies, creates the ground for new markets and fundamental changes. Disruptive innovation transforms the structure of existing markets by offering simpler and lower-cost solutions, and architectural innovation creates new value for the customer by reconfiguring the architecture of products without fundamentally changing their core components (Rothaermel, 2024; Christensen, 2000; Henderson & Clark, 1990).

To analyze these types of innovation, the Innovation Matrix is one of the best-known theoretical frameworks; it is built on the two dimensions of “technology” and “market” and identifies the position of each type of innovation. This matrix helps managers choose an appropriate innovation path given the state of the market and technology; however, on its own, it is not ultimately capable of explaining growth strategies or analyzing customer needs. In the field of organizational growth, the Ansoff Matrix is likewise regarded as one of the most important strategic-planning tools. By combining the product and market dimensions, this matrix introduces four strategies, market penetration, market development, product development, and diversification, and provides a framework for selecting growth pathways for the organization (Ansoff, 1957). Despite its wide use, this model also does not directly take the type of innovation or customer preferences into account.

On the other hand, the Kano Model, by classifying product characteristics into Must-be, One-dimensional, and Attractive features, is an effective tool for analyzing customer needs and satisfaction. It provides the

means to identify and prioritize the characteristics that have the greatest effect on customer satisfaction and has therefore found wide application in new-product design and development. Nevertheless, on its own, the Kano Model is not capable of determining growth strategies or innovation type. Overall, the literature shows that each of these three frameworks explains only part of the decision-making process in innovation management: the Innovation Matrix focuses on the type of innovation, the Ansoff Matrix on the choice of growth strategy, and the Kano Model on identifying and prioritizing customer needs.

A review of the literature shows that both domestic and international studies emphasize the role of multi-criteria decision-making methods, innovation, and understanding of customer needs in improving organizational performance. Among domestic studies, Seyyedi et al. (2024), using simulation and the grey closeness-coefficient TOPSIS method, showed that the use of multi-criteria decision-making models can increase productivity and reduce waste in the home-appliance industry. Similarly, Ghasemi Noghabi and Bahramzadeh (2023) found that creativity in a new product significantly enhances purchase intention through increased perceived value. In international studies, Avikal et al. (2014), by integrating the Kano Model, AHP, and M-TOPSIS, showed that the simultaneous use of these methods increases the precision and comprehensiveness of decision-making in complex problems, and Chen and Lin (2012) likewise confirmed that integrating the Kano Model with TOPSIS improves product evaluation and increases customer satisfaction. Overall, the results of these studies indicate that adopting novel decision-making approaches and simultaneously attending to customer needs and preferences plays an effective role in improving product quality, increasing productivity, strengthening competitive advantage, and enhancing organizational performance, and can provide a suitable basis for the present study.

3. Research Method

In terms of purpose, this research is applied, and in terms of approach, it is qualitative and based on conceptual analysis. In this study, through library studies and a review of the theoretical foundations, the three frameworks of the Innovation Matrix, the Ansoff Matrix, and the Kano Model were analyzed and integrated. Then, by comparing the shared dimensions and concepts of these models, an integrated conceptual framework was presented for explaining innovation and product development strategies. In order to increase the scientific credibility of the proposed framework, a systematic comparison and matching of theoretical foundations and credible scientific sources was used.

4. Findings

A review of the literature and the conceptual integration of the Kano, Ansoff, and Innovation approaches showed that achieving sustainable product and service development requires combining three complementary levels of analysis: the level of customer perception and value creation, the level of choosing a growth strategy, and the level of creating new value through innovation. The findings from the Kano Model analysis showed that customer needs are not uniform in nature in terms of their effect on satisfaction. Must-be features, as basic requirements, act as a precondition for maintaining customer satisfaction; One-dimensional features have a direct relationship with the level of satisfaction, such that improving their performance increases customer satisfaction; whereas Attractive features can be regarded as opportunities

for creating differentiation and new value. Accordingly, the Kano Model provides a basis for identifying and prioritizing customer needs and for determining areas with innovation potential (Brand & Rese, 2026; Shahin, 2014).

Building on this, the integrated findings showed that the information obtained from analyzing customer needs can be used as an input to strategic decision-making in the Ansoff Matrix. This means that once an organization has identified needs with high perceived value, it can in accordance with market conditions and internal capabilities, choose the appropriate growth path from among the four strategies of market penetration, market development, product development, and diversification. The Ansoff Matrix therefore bears the role of translating customer understanding into the organization's strategic direction (Ansoff, 1957).

Likewise, combining the Kano and Ansoff approaches with the concept of innovation showed that effective innovation takes shape when it is guided by customers' real, value-creating needs. In this framework, the Attractive features and some of the One-dimensional features identified in the Kano Model are treated as focal points for innovation and the development of new products or services, whereas Must-be features are more concerned with maintaining baseline quality and preventing customer dissatisfaction.

On this basis, the main finding of the research is the presentation of an integrated conceptual framework in which the Kano Model performs the function of discovering and prioritizing customer needs, the Ansoff Matrix performs the function of determining the growth path, and the Innovation Matrix performs the function of creating and developing new value. This integration shows that organizations can move systematically from understanding customer needs to choosing a growth strategy and then designing innovative solutions, thereby managing product and service development in a systematic manner.

Table 1 Integrated conceptual framework of the Kano Model, the Ansoff Matrix, and Innovation for customer-centered product development

Approach	Main Question	Role in the Proposed Framework	Output
Kano Model	What does the customer consider valuable?	Identifying and prioritizing needs	Must-be, One-dimensional, and Attractive features.
Ansoff Matrix	How should the organization grow?	Selecting the strategic growth path	Market penetration, market development, product development, diversification.
Innovation Matrix	How should new value be created?	Translating customer needs into a new solution	A differentiated product/service.

5. Discussion and Conclusion

The findings of the research showed that integrating the Kano Model, the Ansoff Matrix, and the Innovation Matrix can provide a coherent framework for analyzing and selecting innovation and product development strategies. In this framework, the Kano Model, by identifying and prioritizing customer needs, the Ansoff Matrix, by determining growth pathways, and the Innovation Matrix, by guiding the value-creation process, operate in a complementary manner. This finding is consistent with the prior literature on the importance of customer-centeredness, strategic growth, and innovation in creating competitive advantage; however, by integrating these three approaches, the present study helps to close, to some extent, a gap that exists in the literature. Overall, the proposed framework can serve as a conceptual tool to help managers select appropriate product-development strategies and reduce uncertainty in strategic decision-making. Nevertheless, given the conceptual nature of the research, the empirical validation of this framework across different industries could provide a basis for further development and generalization of the results in future studies.

6. Research Limitations

The most important limitation of this research is its conceptual nature and the lack of empirical validation of the proposed framework in real organizational environments; therefore, generalizing the results to all industries and organizations requires future field studies and empirical testing of the framework in subsequent research.

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